United States Patent Application Publication 20250287309
Kind Code A1
Publication Date September 11, 2025
Inventor(s) DUAN; Weimin et al.

TECHNIQUES TO ACTIVATE SEMI-PERSISTENT SCHEDULING (SPS) OCCASIONS

Abstract

Methods, systems, and devices for wireless communications are described. In some examples, a user equipment (UE) may receive, via a first radio, control signaling that indicates a configuration associated with a wake-up signal (WUS) and that also indicates an association between the WUS and a semi-persistent scheduling (SPS) occasion. As such, the UE may monitor, via a second radio, for the WUS in accordance with the control signaling, where the WUS may indicate whether the SPS occasion includes a physical downlink shared channel (PDSCH) transmission. For example, if the WUS indicates that the SPS occasion includes the PDSCH transmission, the UE may perform a first procedure pertaining to monitoring, via the first radio, the SPS occasion. Alternatively, if the WUS indicates that the SPS occasion is empty, the UE may perform a second procedure pertaining to maintaining the first radio in a sleep mode.

Inventors: **DUAN; Weimin (San Diego, CA), XU; Huilin (Temecula, CA), KHOSHNEVISAN; Mostafa (San Diego, CA), LIU; Le (San Jose, CA), LIU; Kangqi (San Diego, CA), SARKIS; Gabi (San Diego, CA)**

Applicant: **QUALCOMM Incorporated** (San Diego, CA)

Family ID: **94974399**

Appl. No.: **18/596546**

Filed: **March 05, 2024**

Publication Classification

Int. Cl.: **H04W52/02** (20090101); **H04W72/11** (20230101); **H04W72/1273** (20230101); **H04W72/232** (20230101); **H04W72/52** (20230101)

U.S. Cl.:

CPC **H04W52/0235** (20130101); **H04W72/11** (20230101); **H04W72/1273** (20130101); **H04W72/232** (20230101); **H04W72/52** (20230101);

Background/Summary

FIELD OF TECHNOLOGY
[0001] The following relates to wireless communications, including techniques to activate semi-persistent scheduling (SPS) occasions.
BACKGROUND
[0002] Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations, each supporting wireless communication for communication devices, which may be known as user equipment (UE).
SUMMARY
[0003] The described techniques relate to improved methods, systems, devices, and apparatuses that support techniques to activate semi-persistent scheduling (SPS) occasions. For example, the described techniques provide for SPS monitoring indications via a wake-up signal (WUS). In some examples, a UE may receive, via a first radio while operating in an active mode, control signaling indicating a configuration for a low-power WUS, where the control signaling may further indicate the WUS is associated with (e.g., contains information for) one or more SPS occasions. For example, the control signaling may indicate that the WUS is associated with a single SPS occasion. Alternatively, the control signaling may indicate that the WUS is associated with multiple SPS occasions. Accordingly, based on the control signaling, the UE may monitor, via a second radio while the first radio is operating in the sleep mode, for the WUS, where the WUS may indicate whether the associated SPS occasions include a physical downlink shared channel (PDSCH). If the WUS indicates that the associated SPS occasions include a PDSCH, the UE may perform a first procedure to wake-up the first radio, monitor the SPS occasions, and receive the PDSCHs. Alternatively, if the WUS indicates that the associated SPS occasions are empty (e.g., do not include a PDSCH transmission), the UE may maintain the first radio in the sleep mode.
[0004] A method for wireless communications by a UE is described. The method may include receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion, monitoring, via a second radio at the UE, for the WUS in accordance with the configuration, and performing one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.
[0005] A UE for wireless communications is described. The UE may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the UE to receive, via a first radio at the UE, control signaling that

indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion, monitor, via a second radio at the UE, for the WUS in accordance with the configuration, and perform one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0006] Another UE for wireless communications is described. The UE may include means for receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion, means for monitoring, via a second radio at the UE, for the WUS in accordance with the configuration, and means for performing one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0007] A non-transitory computer-readable medium storing code for wireless communications is described. The code may include instructions executable by one or more processors to receive, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion, monitor, via a second radio at the UE, for the WUS in accordance with the configuration, and perform one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0008] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, performing the first procedure may include operations, features, means, or instructions for waking up the first radio of the UE, monitoring, based on waking up the first radio, the SPS occasion, and receiving, via the first radio, the PDSCH based on the monitoring.

[0009] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for monitoring, prior to receiving the PDSCH, for downlink control information (DCI) based on the WUS indicating that the SPS occasion includes the PDSCH, the DCI indicating a set of parameters for receiving the PDSCH.

[0010] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the set of parameters include a quantity of resource blocks (RBs) of the SPS occasion, a modulation and coding scheme (MCS) associated with the PDSCH, a set of resources for indicating feedback associated with the PDSCH, or a combination thereof.

[0011] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for refraining from monitoring for DCI based on the WUS further indicating for the UE to skip monitoring for the DCI, where receiving the PDSCH includes and receiving the PDSCH via the SPS occasion according to a set of parameters indicated in the control signaling.

[0012] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the WUS indicating for the UE to skip monitoring for the DCI may be based on a stability of a channel between the UE and a network entity, a variability of traffic load communicated via the channel, or both.

[0013] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the PDSCH carries data associated with extended reality (XR) applications.

[0014] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the PDSCH may be a retransmission and and the UE monitors for the retransmission or for a DCI that schedules the retransmission based on the WUS indicating that the SPS occasion includes the PDSCH.

[0015] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, performing the second procedure may include operations, features, means, or instructions for refraining from waking up the first radio of the UE, where the first radio may be maintained in the sleep mode based on refraining from waking up the first radio.

[0016] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for refraining from transmitting feedback information associated with the PDSCH based on the WUS indicating that the SPS occasion does not include the PDSCH.

[0017] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for performing the second procedure based on an absence of the WUS during resources associated with the wake-up signal, where the absence of the WUS during the resources indicates that the SPS occasion does not include a PDSCH.

[0018] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for performing the first procedure based on an absence of the WUS during resources associated with the wake-up signal, where the absence of the WUS during the resources indicates that the SPS occasion includes a PDSCH.

[0019] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the control signaling further indicates a jitter range associated with the SPS occasion, and monitoring the WUS may be in accordance with the jitter range associated with the SPS occasion.

[0020] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, performance of the first procedure or the second procedure may be based on bit values of one or more bits of the WUS.

[0021] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, performance of the first procedure or the second procedure may be based on a modulation scheme of the WUS, a data rate of a symbol used to transmit the WUS, a waveform pattern of the WUS, or a combination thereof.

[0022] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, a periodicity of the WUS may be based on a periodicity of the SPS occasion.

[0023] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the WUS may be a low-power WUS.

[0024] A method for wireless communications by an apparatus is described. The method may include transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion and transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0025] An apparatus for wireless communications is described. The apparatus may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the apparatus to transmit control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion and transmit the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0026] Another apparatus for wireless communications is described. The apparatus may include means for transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion and means for transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0027] A non-transitory computer-readable medium storing code for wireless communications is described. The code may include instructions executable by one or more processors to transmit control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion and transmit the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0028] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the WUS indicates that the SPS occasion includes the PDSCH and the method, apparatuses, and non-transitory computer-readable medium may include further operations, features, means, or instructions for transmitting, via the SPS occasion, the PDSCH.

[0029] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, prior to transmitting the PDSCH, DCI that indicates a set of parameters associated with receiving the PDSCH.

[0030] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the set of parameters include a quantity of RBs of the SPS occasion, a MCS associated with the PDSCH, a set of resources for indicating feedback associated with the PDSCH, or a combination thereof.

[0031] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for refraining from transmitting DCI based on the WUS further indicating for the UE to skip monitoring for the DCI.

[0032] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the WUS indicates for the UE to skip monitoring for the DCI may be based on a stability of a channel between the UE and the network entity, a variability of traffic load communicated via the channel, or both.

[0033] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the PDSCH carries data associated with XR applications.

[0034] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the WUS indicates that the SPS occasion does not include the PDSCH and the method, apparatuses, and non-transitory computer-readable medium may include further operations, features, means, or instructions for refraining from transmitting the PDSCH via the SPS occasion.

[0035] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting the WUS according to a periodicity that may be based on a periodicity of the SPS occasion.

[0036] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the control signaling further indicates a jitter range associated with the SPS occasion, and transmitting the WUS may be in accordance with the jitter range associated with the SPS occasion.

[0037] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, bit values of one or more bits of the WUS indicate whether the SPS occasion includes the PDSCH.

[0038] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, a modulation scheme of the WUS, a data rate of a symbol used to transmit the WUS, a waveform pattern of the WUS, or a combination thereof, indicate whether the SPS occasion includes the PDSCH.

[0039] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the WUS may be a low-power WUS.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] FIG. **1** shows an example of a wireless communications system that supports techniques to activate semi-persistent scheduling (SPS) occasions in accordance with one or more aspects of the present disclosure.

[0041] FIG. **2** shows an example of a wireless communications system that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0042] FIG. **3** shows an example of a process flow that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0043] FIGS. **4** and **5** show block diagrams of devices that support techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0044] FIG. **6** shows a block diagram of a communications manager that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0045] FIG. **7** shows a diagram of a system including a device that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0046] FIGS. **8** and **9** show block diagrams of devices that support techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0047] FIG. **10** shows a block diagram of a communications manager that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0048] FIG. **11** shows a diagram of a system including a device that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

[0049] FIGS. **12** through **15** show flowcharts illustrating methods that support techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0050] In some wireless communications system, a network entity may communicate with a user equipment (UE) via semi-persistent scheduling (SPS). For example, a network entity may transmit, via radio resource control (RRC) signaling, a configuration indicating periodic SPS occasions (e.g., periodic time and frequency resources). The network entity may transmit, to the UE, downlink control information activating the SPS configuration, such that the UE may have an indication of when to begin monitoring the SPS occasions. By using the SPS occasions, the network entity may transmit a physical downlink shared channel (PDSCH) to the UE without first transmitting multiple scheduling DCIs.

[0051] In some cases, a main radio of the UE may enter a sleep mode (e.g., an inactive mode), where the UE may power off the main radio and use a low-power wake-up radio (LP-WUR) to monitor for a wake-up signal (WUS). In such cases, if the SPS configuration is activated, then the UE may wake-up the main radio prior to each SPS occasion, monitor the SPS occasion, and receive the PDSCH transmission. In some cases, however, one of the multiple of the periodic SPS occasions may be empty (e.g., not include a PDSCH transmission), which may cause the UE to unnecessarily wake-up the main radio and monitor the SPS occasion, thereby reducing power savings at the UE.

[0052] The techniques described herein may enable the network entity to indicate, via the WUS, whether one or more of the periodic SPS occasions will be empty (e.g., whether one or more of the multiple SPS occasions include a PDSCH transmission), thereby reducing the quantity of times at which the UE wakes up to monitor an empty SPS occasion. For example, the UE may receive, via the main radio, control signaling indicating a configuration associated with the WUS (e.g., one or more time and frequency resources for the WUS, waveform parameters, sequence parameters, or a combination thereof) and also indicating an association between the WUS and a SPS occasion (e.g., that the WUS contains information associated with the SPS occasion). Accordingly, the UE may monitor, via the LP-WUR and while the main radio is operating in the sleep mode, for the WUS, where the WUS may indicate whether the associated SPS occasion includes a PDSCH transmission. For example, the WUS may indicate that the SPS occasion does include a PDSCH transmission and, as such, the UE may wake-up the main radio and monitor the SPS occasion for the PDSCH transmission. Alternatively, the WUS may indicate that the SPS occasion does not include a PDSCH transmission (e.g., is empty) and, as such, the UE may not wake-up the main radio. In this way, the network entity may indicate whether the SPS occasion includes a PDSCH transmission via the WUS.

[0053] Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects are further described in the context of process flow. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to techniques to activate SPS occasions.

[0054] FIG. **1** shows an example of a wireless communications system **100** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The wireless communications system **100** may include one or more devices, such as one or more network devices (e.g., network entities **105**), one or more UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

[0055] The network entities **105** may be dispersed throughout a geographic area to form the wireless communications system **100** and may include devices in different forms or having different capabilities. In various examples, a network entity **105** may be referred to as a network element, a mobility element, a radio access network (RAN) node, or network equipment, among other nomenclature. In some examples, network entities **105** and UEs **115** may wirelessly communicate via communication link(s) **125** (e.g., a radio frequency (RF) access link). For example, a network entity **105** may support a coverage area **110** (e.g., a geographic coverage area) over which the UEs **115** and the network entity **105** may establish the communication link(s) **125**. The coverage area **110** may be an example of a geographic area over which a network entity **105** and a UE **115** may support the communication of signals according to one or more radio access technologies (RATs).

[0056] The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communications system **100**, and each UE **115** may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. **1**. The UEs **115** described herein may be capable of supporting communications with various types of devices in the wireless communications system **100** (e.g., other wireless communication devices, including UEs **115** or network entities **105**), as shown in FIG. **1**.

[0057] As described herein, a node of the wireless communications system **100**, which may be referred to as a network node, or a wireless node, may be a network entity **105** (e.g., any network entity described herein), a UE **115** (e.g., any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE **115**. As another example, a node may be a network entity **105**. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a UE **115**. In another aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a network entity **105**. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE **115**, network entity **105**, apparatus, device, computing system, or the like may include disclosure of the UE **115**, network entity **105**, apparatus, device, computing system, or the like being a node. For example, disclosure that a UE **115** is configured to receive information from a network entity **105** also discloses that a first node is configured to receive information from a second node.

[0058] In some examples, network entities **105** may communicate with a core network **130**, or with one another, or both. For example, network entities **105** may communicate with the core network **130** via backhaul communication link(s) **120** (e.g., in accordance with an S**1**, N**2**, N**3**, or other interface protocol). In some examples, network entities **105** may communicate with one another via backhaul communication link(s) **120** (e.g., in accordance with an X**2**, Xn, or other interface protocol) either directly (e.g., directly between network entities **105**) or indirectly (e.g., via the core network **130**). In some examples, network entities **105** may communicate with one another via a midhaul communication link **162** (e.g., in accordance with a midhaul interface protocol) or a fronthaul communication link **168** (e.g., in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication link(s) **120**, midhaul communication links **162**, or fronthaul communication links **168** may be or include one or more wired links (e.g., an electrical link, an optical fiber link) or one or more wireless links (e.g., a radio link, a wireless optical link), among other examples or various combinations thereof. A UE **115** may communicate with the core network **130** via a communication link **155**.

[0059] One or more of the network entities **105** or network equipment described herein may include or may be referred to as a base station **140** (e.g., a base transceiver station, a radio base station, an NR base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some examples, a network entity **105** (e.g., a base station **140**) may be implemented in an aggregated (e.g., monolithic, standalone) base station architecture, which may be configured to utilize a protocol stack that is physically or logically integrated within one network entity (e.g., a network entity **105** or a single RAN node, such as a base station **140**).

[0060] In some examples, a network entity **105** may be implemented in a disaggregated architecture (e.g., a disaggregated base station architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among multiple network entities (e.g., network entities **105**), such as an integrated access and backhaul (IAB) network, an open RAN (O-RAN) (e.g., a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (e.g., a cloud RAN (C-RAN)). For example, a network entity **105** may include one or more of a central unit (CU), such as a CU **160**, a distributed unit (DU), such as a DU **165**, a radio unit (RU), such as an RU **170**, a RAN Intelligent Controller (RIC), such as an RIC **175** (e.g., a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) system, such as an SMO system **180**, or any combination thereof. An RU **170** may also be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or more components of the network entities **105** in a disaggregated RAN architecture may be co-located, or one or more components of the network entities **105** may be located in distributed locations (e.g., separate physical locations). In some examples, one or more of the network entities **105** of a disaggregated RAN architecture may be implemented as virtual units (e.g., a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

[0061] The split of functionality between a CU **160**, a DU **165**, and an RU **170** is flexible and may support different functionalities depending on which functions (e.g., network layer functions, protocol layer functions, baseband functions, RF functions, or any combinations thereof) are performed at a CU **160**, a DU **165**, or an RU **170**. For example, a functional split of a protocol stack may be employed between a CU **160** and a DU **165** such that the CU **160** may support one or more layers of the protocol stack and the DU **165** may support one or more different layers of the protocol stack. In some examples, the CU **160** may host upper protocol layer (e.g., layer **3** (L**3**), layer **2** (L**2**)) functionality and signaling (e.g., Radio Resource Control (RRC), service data adaptation protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU **160** (e.g., one or more CUs) may be connected to a DU **165** (e.g., one or more DUs) or an RU **170** (e.g., one or more RUs), or some combination thereof, and the DUs **165**, RUs **170**, or both may host lower protocol layers, such as layer **1** (L**1**) (e.g., physical (PHY) layer) or L**2** (e.g., radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU **160**. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU **165** and an RU **170** such that the DU **165** may support one or more layers of the protocol stack and the RU **170** may support one or more different layers of the protocol stack. The DU **165** may support one or multiple different cells (e.g., via one or multiple different RUs, such as an RU **170**). In some cases, a functional split between a CU **160** and a DU **165** or between a DU **165** and an RU **170** may be within a protocol layer (e.g., some functions for a protocol layer may be performed by one of a CU **160**, a DU **165**, or an RU **170**, while other functions of the protocol layer are performed by a different one of the CU **160**, the DU **165**, or the RU **170**). A CU **160** may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU **160** may be connected to a DU **165** via a midhaul communication link **162** (e.g., F**1**, F**1**-c, F**1**-u), and a DU **165** may be connected to an RU **170** via a fronthaul communication link **168** (e.g., open fronthaul (FH) interface). In some examples, a midhaul communication link **162** or a fronthaul communication link **168** may be implemented in accordance with an interface (e.g., a channel) between layers of a protocol stack supported by respective network entities (e.g., one or more of the network entities **105**) that are in communication via such communication links.

[0062] In some wireless communications systems (e.g., the wireless communications system **100**), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (e.g., to a core network **130**). In some cases, in an IAB network, one or more of the network entities **105** (e.g., network entities **105** or IAB node(s) **104**) may be partially controlled by each other. The IAB node(s) **104** may be referred to as a donor entity or an IAB donor. A DU **165** or an RU **170** may be partially controlled by a CU **160** associated with a network entity **105** or base station **140** (such as a donor network entity or a donor base station). The one or more donor entities (e.g., IAB donors) may be in communication with one or more additional devices (e.g., IAB node(s) **104**) via supported access and backhaul links (e.g., backhaul communication link(s) **120**). IAB node(s) **104** may include an IAB mobile termination (IAB-MT) controlled (e.g., scheduled) by one or more DUs (e.g., DUs **165**) of a coupled IAB donor. An IAB-MT may be equipped with an independent set of antennas for relay of communications with UEs **115** or may share the same antennas (e.g., of an RU **170**) of IAB node(s) **104** used for access via the DU **165** of the IAB node(s) **104** (e.g., referred to as virtual IAB-MT (vIAB-MT)). In some examples, the IAB node(s) **104** may include one or more DUs (e.g., DUs **165**) that support communication links with additional entities (e.g., IAB node(s) **104**, UEs **115**) within the relay chain or configuration of the access network (e.g., downstream). In such cases, one or more components of the disaggregated RAN architecture (e.g., the IAB node(s) **104** or components of the IAB node(s) **104**) may be configured to operate according to the techniques described herein.

[0063] In the case of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support test as described herein. For example, some operations described as being performed by a UE **115** or a network entity **105** (e.g., a base station **140**) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (e.g., components such as an IAB node, a DU **165**, a CU **160**, an RU **170**, an RIC **175**, an SMO system **180**).

[0064] A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the "device" may also be referred to as a unit, a station, a terminal, or a client, among other examples.

A UE **115** may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, vehicles, or meters, among other examples.
[0065] The UEs **115** described herein may be able to communicate with various types of devices, such as UEs **115** that may sometimes operate as relays, as well as the network entities **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. **1**.
[0066] The UEs **115** and the network entities **105** may wirelessly communicate with one another via the communication link(s) **125** (e.g., one or more access links) using resources associated with one or more carriers. The term "carrier" may refer to a set of RF spectrum resources having a defined PHY layer structure for supporting the communication link(s) **125**. For example, a carrier used for the communication link(s) **125** may include a portion of an RF spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more PHY layer channels for a given RAT (e.g., LTE, LTE-A, LTE-A Pro, NR). Each PHY layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers. Communication between a network entity **105** and other devices may refer to communication between the devices and any portion (e.g., entity, sub-entity) of a network entity **105**. For example, the terms "transmitting," "receiving," or "communicating," when referring to a network entity **105**, may refer to any portion of a network entity **105** (e.g., a base station **140**, a CU **160**, a DU **165**, a RU **170**) of a RAN communicating with another device (e.g., directly or via one or more other network entities, such as one or more of the network entities **105**).
[0067] Signal waveforms transmitted via a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, in which case the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both), such that a relatively higher quantity of resource elements (e.g., in a transmission duration) and a relatively higher order of a modulation scheme may correspond to a relatively higher rate of communication. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (e.g., a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE **115**.
[0068] The time intervals for the network entities **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of $T_s=1/(\Delta f_{max} \cdot N_f)$ seconds, for which Δf_{max} may represent a supported subcarrier spacing, and N_f may represent a supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).
[0069] Each frame may include multiple consecutively-numbered subframes or slots, and each

subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems, such as the wireless communications system **100**, a slot may further be divided into multiple mini-slots associated with one or more symbols. Excluding the cyclic prefix, each symbol period may be associated with one or more (e.g., N.sub.f) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.
[0070] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs)).
[0071] Physical channels may be multiplexed for communication using a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed for signaling via a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to UEs **115** (e.g., one or more UEs) or may include UE-specific search space sets for sending control information to a UE **115** (e.g., a specific UE).
[0072] The wireless communications system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system **100** may be configured to support ultra-reliable low-latency communications (URLLC). The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.
[0073] In some examples, a UE **115** may be configured to support communicating directly with other UEs (e.g., one or more of the UEs **115**) via a device-to-device (D2D) communication link, such as a D2D communication link **135** (e.g., in accordance with a peer-to-peer (P2P), D2D, or sidelink protocol). In some examples, one or more UEs **115** of a group that are performing D2D communications may be within the coverage area **110** of a network entity **105** (e.g., a base station **140**, an RU **170**), which may support aspects of such D2D communications being configured by (e.g., scheduled by) the network entity **105**. In some examples, one or more UEs **115** of such a group may be outside the coverage area **110** of a network entity **105** or may be otherwise unable to or not configured to receive transmissions from a network entity **105**. In some examples, groups of the UEs **115** communicating via D2D communications may support a one-to-many (1:M) system in

which each UE **115** transmits to one or more of the UEs **115** in the group. In some examples, a network entity **105** may facilitate the scheduling of resources for D2D communications. In some other examples, D2D communications may be carried out between the UEs **115** without an involvement of a network entity **105**.

[0074] The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the network entities **105** (e.g., base stations **140**) associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services **150** for one or more network operators. The IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0075] The wireless communications system **100** may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. Communications using UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than one hundred kilometers) compared to communications using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

[0076] The wireless communications system **100** may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) RAT, or NR technology using an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating using unlicensed RF spectrum bands, devices such as the network entities **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations using unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating using a licensed band (e.g., LAA). Operations using unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0077] A network entity **105** (e.g., a base station **140**, an RU **170**) or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a network entity **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a network entity **105** may be located at diverse geographic locations. A network entity **105** may include an antenna array with a set of rows and columns of antenna ports that the network entity **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may include one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

[0078] Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a network entity **105**, a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating along particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0079] The UEs **115** and the network entities **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) (e.g., including acknowledgements (ACKs) and negative acknowledgements (NACKs)) feedback is one technique for increasing the likelihood that data is received correctly via a communication link (e.g., the communication link(s) **125**, a D2D communication link **135**). HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in relatively poor radio conditions (e.g., low signal-to-noise conditions). In some examples, a device may support same-slot HARQ feedback, in which case the device may provide HARQ feedback in a specific slot for data received via a previous symbol in the slot. In some other examples, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

[0080] In some wireless communications system, the network entity **105** may communicate with the UE **115** via SPS. For example, a network entity **105** may transmit, via RRC signaling, a configuration indicating periodic SPS occasions (e.g., periodic time and frequency resources). The network entity **105** may transmit, to the UE **115**, downlink control information activating the SPS configuration, such that the UE **115** may have an indication of when to begin monitoring the SPS occasions. By using the SPS occasions, the network entity **105** may transmit PDSCH to the UE **115** without first transmitting multiple scheduling DCIs.

[0081] In some cases, a main radio of the UE **115** may enter a sleep mode (e.g., an inactive mode), where the UE **115** may power off the main radio and use a LP-WUR to monitor for a WUS. In such cases, if the SPS configuration is activated, then the UE **115** may wake-up the main radio prior to each SPS occasion, monitor the SPS occasion, and receive the PDSCH transmission. In some cases, however, one of the multiple of the periodic SPS occasions may be empty (e.g., not include a PDSCH transmission), which may cause the UE **115** to unnecessarily wake-up the main radio and monitor the SPS occasion, thereby reducing power savings at the UE **115**.

[0082] The techniques described herein may enable the network entity **105** to indicate, via the WUS, whether one or more of the periodic SPS occasions will be empty (e.g., whether one or more of the multiple SPS occasions include a PDSCH transmission), thereby reducing the quantity of times at which the UE **115** wakes up to monitor an empty SPS occasion. For example, the UE **115** may receive, via the main radio, control signaling indicating a configuration associated the WUS and also indicating an association between the WUS and a SPS occasion (e.g., that the WUS contains information associated with the SPS occasion). Accordingly, the UE **115** may monitor, via the LP-WUR and while the main radio is operating in the sleep mode, for the WUS, where the WUS may indicate whether the associated SPS occasion includes a PDSCH transmission. For example, the WUS may indicate that the SPS occasion does include a PDSCH transmission and, as such, the UE **115** may wake-up the main radio and monitor the SPS occasion for the PDSCH transmission. Alternatively, the WUS may indicate that the SPS occasion does not include a PDSCH transmission (e.g., is empty) and, as such, the UE **115** may not wake-up the main radio. In this way, the network entity **105** may indicate whether the SPS occasion includes a PDSCH transmission via the WUS.

[0083] FIG. **2** shows an example of a wireless communications system **200** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. Aspects of the wireless communications system **200** may implement, or be implemented by, aspects of wireless communications system **100** with reference to FIG. **1**. For example, the wireless communications system **200** may include a network entity **105**-*a* and a UE **115**-*a*, which may be examples of the network entity **105** and the UEs **115** as described herein with reference to FIG. **1**. The techniques described in the context of the wireless communications system **200** may enable the network entity **105**-*a* to indicate, via a WUS **205**, whether one or more SPS occasions **210** include PDSCHs **215**.

[0084] In some cases, the UE **115**-*a* may include one or more radios. For example, the UE **115**-*a* may utilize a main radio **202** during active communications, for example, while the UE **115**-*a* is operating in an active or connected mode. The UE **115**-*a* also may include a LP-WUR **204**, which may be a companion receiver, to the main radio **202**, where the LP-WUR **204** may draw a relatively lower power than the main radio **202** and be used by the UE **115**-*a* to monitor for a WUS **205** while the main radio **202** is operating in a deep sleep state. While the main radio **202** is operating in the sleep mode, the UE **115**-*a*, via the LP-WUR **204**, may monitor for the WUS **205**, such that the UE **115**-*a* may experience power savings during such operations. Accordingly, in response to receiving the WUS **205**, the LP-WUR **204** may wake-up the main radio **202** (e.g., indicate to the main radio **202** to power up) to receive data communications from the network entity **105**-*a*. In such examples, the LP-WUR may consume a relatively lower power than the main radio **202** by design, for example, the LP-WUR may be powered separately than the main radio **202**, include relatively fewer components that consume power than the main radio **202**, or both. By using the LP-WUR **204** while the main radio **202** is in the sleep mode, the UE **115**-*a* may reduce the total power consumption by avoiding unnecessary wake-ups of the main radio **202**, which may increase power consumption at the UE **115**-*a*, and also reduce latency by monitoring WUSs **205**.

[0085] In some examples, to facilitate downlink communication with the UE **115**-*a*, the network entity **105**-*a* may perform semi-persistent scheduling, where the network entity **105**-*a* may configure one or more SPS occasions **210**, which may be periodic resources allocated for downlink transmissions to the UE **115**-*a* (e.g., and retransmissions may be sent via dynamic grants). For example, the network entity **105**-*a* may transmit control signaling **220** (e.g., RRC signaling) that includes an SPS configuration, where the SPS configuration indicates a periodicity of the SPS occasions **210**, a quantity of RBs associated with the SPS occasions **210**, a modulation and coding scheme (MCS) associated with the SPS occasion **210**, physical uplink control channel (PUCCH) resources for HARQ-ACK feedback associated with the SPS occasions **210**, or a combination thereof. That is, the network entity **105**-*a* may indicate, via the control signaling **220**, the same periodicity and parameters for each SPS occasion **210** in the SPS configuration.

[0086] In such examples, the network entity **105**-*a* may activate the SPS configuration indicated by the control signaling **220** via downlink control information (DCI) **230** included in a physical downlink control channel (PDCCH) **225**. In such examples, the DCI **230** may also re-indicate, or update, the SPS configuration, for example, by indicating the time-frequency resources and aforementioned parameters associated with the SPS occasions **210** (in a similar way as dynamic scheduling). Based on receiving the control signaling **220** indicating the SPS configuration and the DCI **230** activating the SPS configuration, the UE **115**-*a* may monitor, via the main radio **202**, each SPS occasion **210** to receive PDSCHs **215**.

[0087] In some cases, however, one of the configured SPS occasions **210** may not carry a PDSCH

215. That is, the network entity **105-*a*** may transmit aperiodic downlink data (e.g., aperiodic PDSCH transmissions) using the SPS occasions **210**, resulting in some SPS occasions **210** being empty (e.g., not carrying a PDSCH **215**). For example, the network entity **105-*a*** may transmit a PDSCH **215-*a*** via the SPS occasion **210-*a*** and transmit a PDSCH **215-*b*** of the SPS occasion **210-*b***, but refrain from transmitting downlink data via the SPS occasion **210-*c***, resulting in the SPS occasion **210-*c***. Such aperiodic downlink transmissions may occur during extended reality (XR) communications, where the periodicity of XR application data may not match with the periodicity of the SPS occasions **210**, where the arrival times of the XR application data may vary due to jitter (e.g., latency) in the communication flow between the XR server and the UE **115-*a***, or both.
[0088] Such empty SPS occasions **210** may lead to a decrease in power savings at the UE **115-*a***. For example, the UE **115-*a*** may receive the WUS **205** indicating for the UE **115-*a*** to wake-up the main radio **202** and monitor for the DCI **230** activating an SPS configuration. However, if one or more of the SPS occasions **210** are empty (e.g., do not include a PDSCH **215**), the UE **115-*a*** may unnecessarily wake-up the main radio **202** and monitor the empty SPS occasion **210-*c***, thereby reducing power savings at the UE **115-*a***. Thus, techniques may be desired to reduce the quantity of times the UE **115-*a*** wakes-up the main radio **202** to monitor empty SPS occasions **210**.
[0089] Further, in some cases, activating the SPS configuration via the DCI **230** may increase the overhead between the UE **115-*a*** and the network entity **105-*a***. For example, if the channel conditions between the UE **115-*a*** and the network entity **105-*a*** remain static (e.g., do not change), the network entity **105-*a*** may unnecessarily update the SPS configuration via the DCI **230** (e.g., update the quantity of RBs of the SPS occasions **210**, update the MCS of the SPS occasions **210**, update the PUCCH resources associated with the SPS occasions **210**). That is, the functionality of scheduling and activating the SPS configuration via the DCI **230** of the PDCCH **225** may be unnecessary under static channel conditions. Thus, techniques may be desired to reduce the overhead between the UE **115-*a*** and the network entity **105-*a*** due to transmitting the DCI **230** activating the SPS configuration.
[0090] The techniques described herein may enable the network entity **105-*a*** to perform SPS activation via the WUS **205**, thereby reducing the overhead between the UE **115-*a*** and the network entity **105-*a***, and also enable the network entity **105-*a*** to indicate, via the WUS **205**, whether one or more SPS occasions **210** include PDSCHs **215** (e.g., whether the SPS occasions **210** are empty), thereby increasing power savings at the UE **115-*a***. For example, if the traffic (e.g., data communication) between the UE **115-*a*** and the network entity **105-*a*** is relatively low (e.g., light) or if the traffic between the UE **115-*a*** and the network entity **105-*a*** includes aperiodic data, such as XR communications, the UE **115-*a*** may skip monitoring one or more SPS occasions **210** that do not carry a PDSCH **215**, skip monitoring the DCI **230** that activates the one or more SPS occasions **210** that do not carry a PDSCH **215**, or both, in response to receiving an indication via the WUS **205**, thereby saving power at the UE **115-*a***.
[0091] In some examples, the network entity **105-*a*** may configure one or more multiple sets of WUSs **205** to be associated with (e.g., be received prior to and contain information for) the SPS occasions **210**. For example, the network entity **105-*a*** may transmit, via the control signaling **220** and in addition to the SPS configuration, a configuration associated with the WUS **205**. In such examples, the network entity **105-*a*** may indicate, via the configuration, time and frequency resources for the WUS **205**, indicate an association between the WUS **205** and the SPS occasions **210**, indicate waveform parameters associated with the WUS **205**, indicate sequence parameters associated with the WUS **205**, or a combination thereof.
[0092] In such examples, the network entity **105-*a*** may set the periodicity of the WUS **205** based on the periodicity indicated in the SPS configuration for the SPS occasions **210**, for example, by setting the periodicity of the WUS **205** to be multiple times that of the periodicity indicated by the SPS configuration for the SPS occasions **210**. Similarly, the network entity **105-*a*** may set the time offset of the WUS **205** based on the SPS configuration for the SPS occasions, for example, by

setting the time offset of the WUS **205** relative to the starting slot of the SPS occasion **210**-*a* (e.g., first SPS occasion **210** of the SPS configuration).
[0093] As an illustrative example, the network entity **105**-*a* may indicate the configuration for the WUS **205** and also indicate that the WUS **205** includes information associated with each SPS occasion **210** (e.g., the SPS occasion **210**-*a*, the SPS occasion **210**-*b*, the SPS occasion **210**-*c*) of an SPS configuration. Alternatively, the network entity **105**-*a* may indicate a configuration for a first WUS **205** and indicate that the first WUS **205** includes information for the SPS occasion **210**-*a*, indicate a configuration for a second WUS **205** and indicate that the second WUS **205** includes information for the SPS occasion **210**-*b* and the SPS occasion **210**-*c*. In some other examples, the network entity **105**-*a* may configure a respective WUS **205** (e.g., transmit respective configurations via the control signaling **220**) for each SPS occasion **210**. In such examples, the network entity **105**-*a* may set the periodicity and time offset of the WUS **205** to be based on those of the SPS configuration of the SPS occasions **210**.
[0094] Accordingly, while the main radio **202** is operating in the sleep mode, the UE **115**-*a* may monitor, using the LP-WUR **204**, for the WUS **205** according to the configuration indicated via the control signaling **220**, where the WUS **205** may provide an indication from the network entity **105**-*a* as to whether one or more SPS occasions **210** (e.g., whether the group of SPS occasions **210** or a single SPS occasion **210**) includes PDSCH **215**. As such, if the WUS **205** indicates that the SPS occasions **210** are empty (e.g., do not include a PDSCH **215**), the UE **115**-*a* may refrain from waking up the main radio **202** and maintain the main radio **202** in the sleep mode, thereby refraining from using the main radio **202** to monitor for the DCI **230** and empty SPS occasions **210**. Alternatively, if the WUS **205** indicates that one or more of the SPS occasions are not empty include a PDSCH **215**, the UE **115**-*a* may wake-up the main radio **202** to monitor for the DCI **230**, receive the PDSCH **215**, and decode the PDSCH **215**.
[0095] As an illustrative example, the network entity **105**-*a* may indicate, via the control signaling **220**, that the WUS **205** is associated with the SPS occasion **210**-*a*, the SPS occasion **210**-*b*, and the SPS occasion **210**-*c*. Accordingly, the UE **115**-*a* may receive, via the LP-WUR **204**, the WUS **205**, where the WUS **205** may indicate that the SPS occasion **210**-*a* includes the PDSCH **215**-*a*, indicate that the SPS occasion **210**-*b* includes the PDSCH **215**-*b*, and indicate that the SPS occasion **210**-*c* does not include a PDSCH **215** (e.g., is empty). Based on the WUS **205**, the UE **115**-*a* may wake-up the main radio **202**, monitor for the PDCCH for the DCI **230** activating the SPS configuration associated with the SPS occasions **210**, monitor the SPS occasion **210**-*a* for the PDSCH **215**-*a* and the SPS occasion **210**-*b* for the PDSCH **215**-*b*, and refrain from monitoring the SPS occasion **210**-*c*.
[0096] In some examples, if the WUS **205** indicates that the one or more SPS occasions **210** are not empty (e.g., include a PDSCH **215**), the network entity **105**-*a* may further indicate, via the WUS **205**, whether the UE **115**-*a* is to wake-up the main radio **202** and monitor for the DCI **230**. That is, because the network entity **105**-*a* indicates that the one or more SPS occasions **210** include a PDSCH **215** via the WUS **205**, the UE **115**-*a* may refrain from receiving the activation information via the DCI **230**, and accordingly, skip monitoring for the DCI **230** in order to further save power.
[0097] For example, if the channel between the UE **115**-*a* and the network entity **105**-*a* is stable (e.g., static, not changing) and the traffic load is static (e.g., consistent, not variable), the network entity **105**-*a* may refrain from dynamically updating the parameters of the SPS occasions **210** (e.g., quantity of RBs for each SPS occasion **210**, MCS associated with each SPS occasion, PUCCH resources for HARQ-ACK feedback associated with each SPS occasion **210**). Accordingly, the network entity **105**-*a* may transmit the WUS **205** to further indicate that the UE **115**-*a* is to skip monitoring the DCI **230** and directly monitor the SPS occasions **210** to receive and decode the PDSCHs **215** using the previous scheduling information (e.g., SPS configuration indicated via the control signaling **220** or previous DCI **230**), which may further save power at the UE **115**-*a*.
[0098] As an illustrative example, the network entity **105**-*a* may indicate, via the control signaling

220, that the WUS **205** is associated with the SPS occasion **210**-*a*, the SPS occasion **210**-*b*, and the SPS occasion **210**-*c*. Accordingly, the UE **115**-*a* may receive, via the LP-WUR **204**, the WUS **205**, where the WUS **205** may indicate that the SPS occasion **210**-*a* includes the PDSCH **215**-*a*, indicate that the SPS occasion **210**-*b* includes the PDSCH **215**-*b*, and indicate that the SPS occasion **210**-*c* does not include a PDSCH **215** (e.g., is empty). The WUS **205** may further indicate to skip monitoring the PDCCH for the DCI **230**. Based on the WUS **205**, the UE **115**-*a* may wake-up the main radio **202**, refrain from monitoring the PDCCH for the DCI **230**, monitor the SPS occasion **210**-*a* for the PDSCH **215**-*a* and the SPS occasion **210**-*b* for the PDSCH **215**-*b*, and refrain from monitoring the SPS occasion **210**-*c*.

[0099] In some examples, if the UE **115**-*a* does not detect the WUS **205** (e.g., does not receive the WUS **205**), the UE **115**-*a* may refrain from waking up the main radio **202** and skip monitoring the SPS occasions **210**. For example, the network entity **105**-*a* may indicate that each of the SPS occasions **210** do not include a PDSCH **215** (e.g., are empty) based on not transmitting the WUS **205**. Accordingly, the UE **115**-*a* may not receive the WUS **205** and skip monitoring the SPS occasions. Alternatively, if the UE **115**-*a* does not detect the WUS **205**, the UE **115**-*a* may wake-up the main radio **202**, monitor the SPS occasions **210**, and receive the PDSCHs **215**. For example, if the WUS **205** is not transmitted from the network entity **105**-*a*, the UE **115**-*a* may determine that the PDSCHs **215** (e.g., downlink data) is periodic and present in the SPS occasions **210**. Accordingly, the UE **115**-*a* may wakeup the main radio **202**, monitor the SPS occasions **210**, and receive the PDSCHs **215** without first monitoring for the WUS **205**.

[0100] The UE **115**-*a* may determine whether the SPS occasions **210** includes a PDSCH **215** according to one or more bits of the WUS **205**, according to a waveform of the WUS **205**, according to a data rate of the WUS **205**, or a combination thereof. In some examples, the WUS **205** may include one or more bits indicating whether the SPS occasions **210** include a PDSCH **215**, where such bits may be the least significant bits (LSBs) of the WUS **205**, while legacy information for wake-up procedures may be the most significant bits (MSBs) of the WUS **205**, or vice versa. As an illustrative example, the WUS **205** may include three bits (e.g., three LSBs), where a first of the three bits may correspond to the SPS occasion **210**-*a*, a second of the three bits may correspond to the SPS occasion **210**-*b*, and a third of the three bits may correspond to the SPS occasion **210**-*c*. Accordingly, the UE **115**-*a* may determine whether each of the SPS occasions **210** include a PDSCH **215** based on the bit values of the corresponding bits (e.g., a bit value of '1' indicates not empty, a bit value of '0' indicates empty, or vice versa). In this way, the network entity **105**-*a* may indicate whether the SPS occasions are empty according to bits added to the WUS **205**.

[0101] In some other examples, the network entity **105**-*a* may transmit the WUS **205** according to one or more data rates (e.g., symbol rates) to indicate whether the SPS occasions **210** include a PDSCH **215**. For example, several data rates (e.g., symbol rates) may be defined, such as a default data rate, a first data rate, a second data rate, a third data rate, and so on, where each data rate may correspond to a respective WUS configuration. Each WUS configuration may provide an indication as to which SPS occasion **210** is empty or not. Accordingly, if the WUS **205** is transmitted according to the default data rate (e.g., corresponding to the default WUS configuration), the UE **115**-*a* may determine that each of the SPS occasions **210** include a PDSCH **215** or do not include a PDSCH **215**. As such, if the UE **115**-*a* does not detect a WUS **205** with the default data rate, the UE **115**-*a* may have an indication that each status (e.g., whether empty or not) of the SPS occasion **210** is different. For example, if the UE **115**-*a* detects that the WUS **205** has a non-default data rate, the UE **115**-*a* may identify the associated non-default WUS configuration, which may indicate whether a specific SPS occasion **210** is empty or not.

[0102] In some other examples, the network entity **105**-*a* may transmit the WUS **205** with a hybrid waveform (e.g., a first portion of the waveform is different from a second portion of the waveform), such as an on-off keying (OOK) and frequency modulated continuous wave (FMCW) waveform, a frequency shift keying (FSK) and FMCW waveform, an OOK waveform with an

FMCW slope, or an FSK waveform with an FMCW slope. To produce such waveforms, the network entity **105**-*a* may modulate the WUS **205** according to a combination of MCSs (e.g., to produce the OOK FMCW waveform, the network entity **105**-*a* may apply an OOK MCS and FMCW MCS to transmit the WUS **205**). Accordingly, the UE **115**-*a* may determine whether one or more SPS occasions **210** include a PDSCH **215** based on the hybrid waveform of the WUS **205**. For example, if the UE **115**-*a* detects a first waveform (e.g., a default waveform), then the UE **115**-*a* may determine that each of the SPS occasions **210** include a PDSCH **215** or do not include a PDSCH **215**. Alternatively, if the UE **115**-*a* does not detect a WUS **205** (e.g., a default waveform), the UE **115**-*a* may combine the first waveform with a second waveform to form the hybrid waveform and detect whether a specific SPS occasion **210** is empty or not based on the hybrid waveform. In such examples, if the UE **115**-*a* does not detect a defined waveform for the WUS **205**, the UE **115**-*a* may refrain from waking up the main radio **202** and skip monitoring the SPS occasions or wake-up the main radio **202** and monitor the SPS occasions.

[0103] As described herein, the network entity **105**-*a* may use the SPS occasions **210** to transmit XR traffic, however, the jitter associated with such XR traffic (e.g., PDSCHs **215** carrying XR data) may result in one or more of the SPS occasions **210** being empty. Accordingly, the network entity **105**-*a* may configure one or more WUSs **205** around the SPS occasions **210** carrying the XR traffic, where the WUS **205** may enable high granularity wake-up of the main radio **202** to achieve low latency, while maintain low power consumption with LP-WUR **204**. As such, the network entity **105**-*a* may use the WUS **205** to indicate whether one or more SPS occasions **210** carrying XR traffic or not (e.g., whether the SPS occasions are empty). Further, to reduce power consumption at the UE **115**-*a*, the network entity **105**-*a* may transmit, via the control signaling **220**, jitter range information (e.g., average arrival time of XR traffic relative to a delivery deadline, a time period prior to the average arrival time, a time period after the average arrival time) associated with the XR traffic, such that the UE **115**-*a* may monitor for WUSs **205** that are within the jitter range of the SPS occasions **210**.

[0104] In some examples, to reduce unnecessary HARQ-ACK transmissions from the UE **115**-*a*, the UE **115**-*a* may refrain from transmitting HARQ information based on the WUS **205**. For example, if the WUS **205** indicates that the SPS occasions **210** do not include a PDSCH **215**, then the UE **115**-*a* may refrain from transmitting a NACK for the associated SPS occasions, at least if the HARQ-ACK information is the singular uplink transmission (e.g., only uplink transmission). Otherwise, the UE **115**-*a* may utilize the PUCCH resources in the SPS configuration for HARQ-ACK transmissions when the SPS occasions **210** include a PDSCH **215**.

[0105] In some other examples, the UE **115**-*a* may determine whether to monitor for a retransmission of a PDSCH **215** based on the WUS **205**. For example, the UE **115**-*a* may determine whether to monitor for a retransmission of a PDSCH **215** original scheduled in the SPS occasions **210** based on whether the WUS **205** indicates the presence or absence of the PDSCH in the SPS occasions **210**. If the WUS **205** indicates that the SPS occasions **210** are empty, then the UE **115**-*a* may refrain from monitoring for retransmissions. Otherwise, if the UE **115**-*a* receives an indication that the SPS occasions **210** include a PDSCH **215**, the UE **115**-*a* may also receive an indication in the WUS **205** as to whether the UE **115**-*a* is to monitor for retransmissions of the PDSCH **215**. For example, the WUS **205** may indicate for the UE **115**-*a* to receive the PDSCH retransmission, indicate for the UE **115**-*a* to receive the DCI **230** scheduling the PDSCH retransmission, or both.

[0106] FIG. **3** shows an example of a process flow **300** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. Aspects of the process flow **300** may implement, or be implemented by, aspects of wireless communications system **100** and the wireless communications system **200**, as described herein with reference to FIGS. **1** and **2**. For example, the process flow **300** may include a network entity **105**-*b* and a UE **115**-*b*, which may be examples of the network entity **105** and the UEs **115** as described herein with reference to FIGS. **1** and **2**. The techniques described in the context of the process flow **300** may enable the network entity **105**-*b* to indicate, via a WUS, whether one or more SPS occasions include PDSCH transmissions.

[0107] At **305**, the UE **115**-*b* may receive, via a first radio (e.g., the main radio **202**), control signaling (e.g., the control signaling **220**) indicating a configuration associated with a WUS (e.g., the WUS **205**) and indicating an association between the WUS and a SPS occasion (e.g., SPS occasion **210**). For example, the UE **115**-*b* may receive the control signaling in accordance with the techniques described herein with reference to FIG. **2**. In such examples, the configuration associated with the WUS may include time and frequency resources associated with the WUS, waveform parameters associated with the WUS (e.g., waveform type, MCS, data rate, or a combination thereof), sequence parameters associated with the WUS (e.g., a quantity of bits associated with the WUS), a data rate associated with the WUS, or any combination thereof.

[0108] At **310**, the UE **115**-*b* may monitor for, and receive, via a second radio (e.g., LP-WUR **204**), the WUS, where the WUS indicates whether the SPS occasion includes a PDSCH (e.g., PDSCH **215**). The network entity **105**-*b* may indicate, via the WUS, whether the SPS occasion includes a PDSCH via one or more bits, via a data rate of a symbol of the WUS, a waveform of the WUS, or a combination thereof, as described herein with reference to FIG. **2**. Accordingly, the UE **115**-*b* may perform one of a first procedure (e.g., wakeup the main radio and receive PDSCH) or a second procedure (e.g., maintain the main radio in the sleep mode) based on the indication in the WUS.

[0109] At **315**, if the WUS indicates that the SPS occasion does not include a PDSCH (e.g., is empty), the UE **115**-*b* may maintain the main radio in the sleep mode and refrain from waking-up the first radio. Alternatively, at **320**, if the WUS indicates that the SPS occasion does include a PDSCH, the UE **115**-*b* may wake-up the first radio. At **325**, the UE **115**-*b* may monitor the SPS occasion and receive the PDSCH. In some examples, prior to monitoring for the SPS occasion, the UE **115**-*b* may monitor a PDCCH for a DCI (e.g., DCI **230**) that activates the SPS occasion, where receiving the PDSCH is based on receiving the DCI. Alternatively, the WUS may indicate for the UE **115**-*b* to skip monitoring the DCI, and instead monitor the SPS occasion directly to receive the PDSCH.

[0110] FIG. **4** shows a block diagram **400** of a device **405** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The device **405** may be an example of aspects of a UE **115** as described herein. The device **405** may include a receiver **410**, a transmitter **415**, and a communications manager **420**. The device **405**, or one or more components of the device **405** (e.g., the receiver **410**, the transmitter **415**, the communications manager **420**), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0111] The receiver **410** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to techniques to activate SPS occasions). Information may be passed on to other components of the device **405**. The receiver **410** may utilize a single antenna or a set of multiple antennas.

[0112] The transmitter **415** may provide a means for transmitting signals generated by other components of the device **405**. For example, the transmitter **415** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to techniques to activate SPS occasions). In some examples, the transmitter **415** may be co-located with a receiver **410** in a transceiver module. The transmitter **415** may utilize a single antenna or a set of multiple antennas.

[0113] The communications manager **420**, the receiver **410**, the transmitter **415**, or various combinations or components thereof may be examples of means for performing various aspects of techniques to activate SPS occasions as described herein. For example, the communications manager **420**, the receiver **410**, the transmitter **415**, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0114] In some examples, the communications manager **420**, the receiver **410**, the transmitter **415**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include at least one of a processor, a digital signal processor (DSP), a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure. In some examples, at least one processor and at least one memory coupled with the at least one processor may be configured to perform one or more of the functions described herein (e.g., by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0115] Additionally, or alternatively, the communications manager **420**, the receiver **410**, the transmitter **415**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by at least one processor (e.g., referred to as a processor-executable code). If implemented in code executed by at least one processor, the functions of the communications manager **420**, the receiver **410**, the transmitter **415**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).

[0116] In some examples, the communications manager **420** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **410**, the transmitter **415**, or both. For example, the communications manager **420** may receive information from the receiver **410**, send information to the transmitter **415**, or be integrated in combination with the receiver **410**, the transmitter **415**, or both to obtain information, output information, or perform various other operations as described herein.

[0117] The communications manager **420** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **420** is capable of, configured to, or operable to support a means for receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The communications manager **420** is capable of, configured to, or operable to support a means for monitoring, via a second radio at the UE, for the WUS in accordance with the configuration. The communications manager **420** is capable of, configured to, or operable to support a means for performing one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0118] By including or configuring the communications manager **420** in accordance with examples as described herein, the device **405** (e.g., at least one processor controlling or otherwise coupled with the receiver **410**, the transmitter **415**, the communications manager **420**, or a combination thereof) may support techniques for indicating empty SPS occasions via a WUS, which may provide for reduced processing, reduced power consumption, and a more efficient utilization of communication resources.

[0119] FIG. **5** shows a block diagram **500** of a device **505** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The device **505** may be an example of aspects of a device **405** or a UE **115** as described herein. The device **505** may include a receiver **510**, a transmitter **515**, and a communications manager **520**. The device **505**, or one or more components of the device **505** (e.g., the receiver **510**, the transmitter **515**, the communications manager **520**), may include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0120] The receiver **510** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to techniques to activate SPS occasions). Information may be passed on to other components of the device **505**. The receiver **510** may utilize a single antenna or a set of multiple antennas.

[0121] The transmitter **515** may provide a means for transmitting signals generated by other components of the device **505**. For example, the transmitter **515** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to techniques to activate SPS occasions). In some examples, the transmitter **515** may be co-located with a receiver **510** in a transceiver module. The transmitter **515** may utilize a single antenna or a set of multiple antennas.

[0122] The device **505**, or various components thereof, may be an example of means for performing various aspects of techniques to activate SPS occasions as described herein. For example, the communications manager **520** may include a control signaling component **525**, a WUR component **530**, an SPS procedure component **535**, or any combination thereof. The communications manager **520** may be an example of aspects of a communications manager **420** as described herein. In some examples, the communications manager **520**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **510**, the transmitter **515**, or both. For example, the communications manager **520** may receive information from the receiver **510**, send information to the transmitter **515**, or be integrated in combination with the receiver **510**, the transmitter **515**, or both to obtain information, output information, or perform various other operations as described herein.

[0123] The communications manager **520** may support wireless communications in accordance with examples as disclosed herein. The control signaling component **525** is capable of, configured to, or operable to support a means for receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The WUR component **530** is capable of, configured to, or operable to support a means for monitoring, via a second radio at the UE, for the WUS in accordance with the configuration. The SPS procedure component **535** is capable of, configured to, or operable to support a means for performing one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0124] FIG. **6** shows a block diagram **600** of a communications manager **620** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The communications manager **620** may be an example of aspects of a communications manager **420**, a communications manager **520**, or both, as described herein. The communications manager **620**, or various components thereof, may be an example of means for performing various aspects of techniques to activate SPS occasions as described herein. For example, the communications manager **620** may include a control signaling component **625**, a WUR component **630**, an SPS procedure component **635**, a main radio component **640**, an SPS monitoring component **645**, an PDSCH reception component **650**, a DCI reception component **655**, an HARQ-ACK component **660**, or any combination thereof. Each of these components, or components or subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0125] The communications manager **620** may support wireless communications in accordance with examples as disclosed herein. The control signaling component **625** is capable of, configured to, or operable to support a means for receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The WUR component **630** is capable of, configured to, or operable to support a means for monitoring, via a second radio at the UE, for the WUS in accordance with the configuration. The SPS procedure component **635** is capable of, configured to, or operable to support a means for performing one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0126] In some examples, to support performing the first procedure, the main radio component **640** is capable of, configured to, or operable to support a means for waking up the first radio of the UE. In some examples, to support performing the first procedure, the SPS monitoring component **645** is capable of, configured to, or operable to support a means for monitoring, based on waking up the first radio, the SPS occasion. In some examples, to support performing the first procedure, the PDSCH reception component **650** is capable of, configured to, or operable to support a means for receiving, via the first radio, the PDSCH based on the monitoring.

[0127] In some examples, the DCI reception component **655** is capable of, configured to, or operable to support a means for monitoring, prior to receiving the PDSCH, for DCI based on the WUS indicating that the SPS occasion includes the PDSCH, the DCI indicating a set of parameters for receiving the PDSCH.

[0128] In some examples, the set of parameters include a quantity of RBs of the SPS occasion, a MCS associated with the PDSCH, a set of resources for indicating feedback associated with the PDSCH, or a combination thereof.

[0129] In some examples, the DCI reception component **655** is capable of, configured to, or operable to support a means for refraining from monitoring for DCI based on the WUS further indicating for the UE to skip monitoring for the DCI. In some examples, the PDSCH reception component **650** is capable of, configured to, or operable to support a means for receiving the PDSCH via the SPS occasion according to a set of parameters indicated in the control signaling.

[0130] In some examples, the WUS indicating for the UE to skip monitoring for the DCI is based on a stability of a channel between the UE and a network entity, a variability of traffic load communicated via the channel, or both.

[0131] In some examples, the PDSCH carries data associated with extended reality applications.

[0132] In some examples, the PDSCH is a retransmission and. In some examples, the UE monitors for the retransmission or for a DCI that schedules the retransmission based on the WUS indicating that the SPS occasion includes the PDSCH.

[0133] In some examples, to support performing the second procedure, the main radio component **640** is capable of, configured to, or operable to support a means for refraining from waking up the first radio of the UE, where the first radio is maintained in the sleep mode based on refraining from waking up the first radio.

[0134] In some examples, the HARQ-ACK component **660** is capable of, configured to, or operable to support a means for refraining from transmitting feedback information associated with the PDSCH based on the WUS indicating that the SPS occasion does not include the PDSCH.

[0135] In some examples, the WUR component **630** is capable of, configured to, or operable to support a means for performing the second procedure based on an absence of the WUS during resources associated with the WUS, where the absence of the WUS during the resources indicates that the SPS occasion does not include a PDSCH.

[0136] In some examples, the WUR component **630** is capable of, configured to, or operable to support a means for performing the first procedure based on an absence of the WUS during resources associated with the WUS, where the absence of the WUS during the resources indicates that the SPS occasion includes a PDSCH.

[0137] In some examples, the control signaling further indicates a jitter range associated with the SPS occasion, and monitoring the WUS is in accordance with the jitter range associated with the SPS occasion.
[0138] In some examples, performance of the first procedure or the second procedure is based on bit values of one or more bits of the WUS.
[0139] In some examples, performance of the first procedure or the second procedure is based on a modulation scheme of the WUS, a data rate of a symbol used to transmit the WUS, a waveform pattern of the WUS, or a combination thereof.
[0140] In some examples, a periodicity of the WUS is based on a periodicity of the SPS occasion.
[0141] In some examples, the WUS is a low-power WUS.
[0142] FIG. **7** shows a diagram of a system **700** including a device **705** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The device **705** may be an example of or include components of a device **405**, a device **505**, or a UE **115** as described herein. The device **705** may communicate (e.g., wirelessly) with one or more other devices (e.g., network entities **105**, UEs **115**, or a combination thereof). The device **705** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **720**, an input/output (I/O) controller, such as an I/O controller **710**, a transceiver **715**, one or more antennas **725**, at least one memory **730**, code **735**, and at least one processor **740**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **745**).
[0143] The I/O controller **710** may manage input and output signals for the device **705**. The I/O controller **710** may also manage peripherals not integrated into the device **705**. In some cases, the I/O controller **710** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **710** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **710** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **710** may be implemented as part of one or more processors, such as the at least one processor **740**. In some cases, a user may interact with the device **705** via the I/O controller **710** or via hardware components controlled by the I/O controller **710**.
[0144] In some cases, the device **705** may include a single antenna. However, in some other cases, the device **705** may have more than one antenna, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **715** may communicate bi-directionally via the one or more antennas **725** using wired or wireless links as described herein. For example, the transceiver **715** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **715** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **725** for transmission, and to demodulate packets received from the one or more antennas **725**. The transceiver **715**, or the transceiver **715** and one or more antennas **725**, may be an example of a transmitter **415**, a transmitter **515**, a receiver **410**, a receiver **510**, or any combination thereof or component thereof, as described herein.
[0145] The at least one memory **730** may include random access memory (RAM) and read-only memory (ROM). The at least one memory **730** may store computer-readable, computer-executable, or processor-executable code, such as the code **735**. The code **735** may include instructions that, when executed by the at least one processor **740**, cause the device **705** to perform various functions described herein. The code **735** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **735** may not be directly executable by the at least one processor **740** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the at least one memory **730** may

include, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

[0146] The at least one processor **740** may include one or more intelligent hardware devices (e.g., one or more general-purpose processors, one or more DSPs, one or more central processing units (CPUs), one or more graphics processing units (GPUs), one or more neural processing units (NPUs) (also referred to as neural network processors or deep learning processors (DLPs)), one or more microcontrollers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some cases, the at least one processor **740** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the at least one processor **740**. The at least one processor **740** may be configured to execute computer-readable instructions stored in a memory (e.g., the at least one memory **730**) to cause the device **705** to perform various functions (e.g., functions or tasks supporting techniques to activate SPS occasions). For example, the device **705** or a component of the device **705** may include at least one processor **740** and at least one memory **730** coupled with or to the at least one processor **740**, the at least one processor **740** and the at least one memory **730** configured to perform various functions described herein. In some examples, the at least one processor **740** may include multiple processors and the at least one memory **730** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions described herein. In some examples, the at least one processor **740** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **740**) and memory circuitry (which may include the at least one memory **730**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **740** or a processing system including the at least one processor **740** may be configured to, configurable to, or operable to cause the device **705** to perform one or more of the functions described herein. Further, as described herein, being "configured to," being "configurable to," and being "operable to" may be used interchangeably and may be associated with a capability, when executing code **735** (e.g., processor-executable code) stored in the at least one memory **730** or otherwise, to perform one or more of the functions described herein.

[0147] The communications manager **720** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **720** is capable of, configured to, or operable to support a means for receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The communications manager **720** is capable of, configured to, or operable to support a means for monitoring, via a second radio at the UE, for the WUS in accordance with the configuration. The communications manager **720** is capable of, configured to, or operable to support a means for performing one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0148] By including or configuring the communications manager **720** in accordance with examples as described herein, the device **705** may support techniques for indicating empty SPS occasions via a WUS, which may provide for reduced processing, reduced power consumption, and a more efficient utilization of communication resources.

[0149] In some examples, the communications manager **720** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **715**, the one or more antennas **725**, or any combination thereof. Although the communications manager **720** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **720** may be supported by or performed by the at least one processor **740**, the at least one memory **730**, the code **735**, or any combination thereof. For example, the code **735** may include instructions executable by the at least one processor **740** to cause the device **705** to perform various aspects of techniques to activate SPS occasions as described herein, or the at least one processor **740** and the at least one memory **730** may be otherwise configured to, individually or collectively, perform or support such operations.

[0150] FIG. **8** shows a block diagram **800** of a device **805** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The device **805** may be an example of aspects of a network entity **105** as described herein. The device **805** may include a receiver **810**, a transmitter **815**, and a communications manager **820**. The device **805**, or one or more components of the device **805** (e.g., the receiver **810**, the transmitter **815**, the communications manager **820**), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0151] The receiver **810** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **805**. In some examples, the receiver **810** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **810** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

[0152] The transmitter **815** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **805**. For example, the transmitter **815** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **815** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **815** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **815** and the receiver **810** may be co-located in a transceiver, which may include or be coupled with a modem.

[0153] The communications manager **820**, the receiver **810**, the transmitter **815**, or various combinations or components thereof may be examples of means for performing various aspects of techniques to activate SPS occasions as described herein. For example, the communications manager **820**, the receiver **810**, the transmitter **815**, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0154] In some examples, the communications manager **820**, the receiver **810**, the transmitter **815**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include at least one of a processor, a DSP, a CPU, an ASIC, an FPGA or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure. In some examples, at least one processor and at least one memory coupled with the at least one processor may be configured to perform one or more of the functions described herein (e.g., by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0155] Additionally, or alternatively, the communications manager **820**, the receiver **810**, the transmitter **815**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by at least one processor (e.g., referred to as a processor-executable code). If implemented in code executed by at least one processor, the functions of the communications manager **820**, the receiver **810**, the transmitter **815**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).
[0156] In some examples, the communications manager **820** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **810**, the transmitter **815**, or both. For example, the communications manager **820** may receive information from the receiver **810**, send information to the transmitter **815**, or be integrated in combination with the receiver **810**, the transmitter **815**, or both to obtain information, output information, or perform various other operations as described herein.
[0157] The communications manager **820** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **820** is capable of, configured to, or operable to support a means for transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The communications manager **820** is capable of, configured to, or operable to support a means for transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.
[0158] By including or configuring the communications manager **820** in accordance with examples as described herein, the device **805** (e.g., at least one processor controlling or otherwise coupled with the receiver **810**, the transmitter **815**, the communications manager **820**, or a combination thereof) may support techniques for indicating empty SPS occasions via a WUS, which may provide for reduced processing, reduced power consumption, and a more efficient utilization of communication resources.
[0159] FIG. **9** shows a block diagram **900** of a device **905** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The device **905** may be an example of aspects of a device **805** or a network entity **105** as described herein. The device **905** may include a receiver **910**, a transmitter **915**, and a communications manager **920**. The device **905**, or one or more components of the device **905** (e.g., the receiver **910**, the transmitter **915**, the communications manager **920**), may include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).
[0160] The receiver **910** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **905**. In some examples, the receiver **910** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **910** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.
[0161] The transmitter **915** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **905**. For example, the transmitter **915** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels,

channels associated with a protocol stack). In some examples, the transmitter **915** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **915** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **915** and the receiver **910** may be co-located in a transceiver, which may include or be coupled with a modem.

[0162] The device **905**, or various components thereof, may be an example of means for performing various aspects of techniques to activate SPS occasions as described herein. For example, the communications manager **920** may include a scheduling component **925** a WUS transmission component **930**, or any combination thereof. The communications manager **920** may be an example of aspects of a communications manager **820** as described herein. In some examples, the communications manager **920**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **910**, the transmitter **915**, or both. For example, the communications manager **920** may receive information from the receiver **910**, send information to the transmitter **915**, or be integrated in combination with the receiver **910**, the transmitter **915**, or both to obtain information, output information, or perform various other operations as described herein.

[0163] The communications manager **920** may support wireless communications in accordance with examples as disclosed herein. The scheduling component **925** is capable of, configured to, or operable to support a means for transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The WUS transmission component **930** is capable of, configured to, or operable to support a means for transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0164] FIG. **10** shows a block diagram **1000** of a communications manager **1020** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The communications manager **1020** may be an example of aspects of a communications manager **820**, a communications manager **920**, or both, as described herein. The communications manager **1020**, or various components thereof, may be an example of means for performing various aspects of techniques to activate SPS occasions as described herein. For example, the communications manager **1020** may include a scheduling component **1025**, a WUS transmission component **1030**, an PDSCH transmission component **1035**, a DCI component **1040**, or any combination thereof. Each of these components, or components or subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses). The communications may include communications within a protocol layer of a protocol stack, communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack, within a device, component, or virtualized component associated with a network entity **105**, between devices, components, or virtualized components associated with a network entity **105**), or any combination thereof.

[0165] The communications manager **1020** may support wireless communications in accordance with examples as disclosed herein. The scheduling component **1025** is capable of, configured to, or operable to support a means for transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The WUS transmission component **1030** is capable of, configured to, or operable to support a means for transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0166] In some examples, the WUS indicates that the SPS occasion includes the PDSCH, and the PDSCH transmission component **1035** is capable of, configured to, or operable to support a means for transmitting, via the SPS occasion, the PDSCH.

[0167] In some examples, the DCI component **1040** is capable of, configured to, or operable to support a means for transmitting, prior to transmitting the PDSCH, DCI that indicates a set of parameters associated with receiving the PDSCH.
[0168] In some examples, the set of parameters include a quantity of RBs of the SPS occasion, a MCS associated with the PDSCH, a set of resources for indicating feedback associated with the PDSCH, or a combination thereof.
[0169] In some examples, the DCI component **1040** is capable of, configured to, or operable to support a means for refraining from transmitting DCI based on the WUS further indicating for the UE to skip monitoring for the DCI.
[0170] In some examples, the WUS indicates for the UE to skip monitoring for the DCI is based on a stability of a channel between the UE and the network entity, a variability of traffic load communicated via the channel, or both.
[0171] In some examples, the PDSCH carries data associated with extended reality applications.
[0172] In some examples, the WUS indicates that the SPS occasion does not include the PDSCH, and the PDSCH transmission component **1035** is capable of, configured to, or operable to support a means for refraining from transmitting the PDSCH via the SPS occasion.
[0173] In some examples, the WUS transmission component **1030** is capable of, configured to, or operable to support a means for transmitting the WUS according to a periodicity that is based on a periodicity of the SPS occasion.
[0174] In some examples, the control signaling further indicates a jitter range associated with the SPS occasion, and transmitting the WUS is in accordance with the jitter range associated with the SPS occasion.
[0175] In some examples, bit values of one or more bits of the WUS indicate whether the SPS occasion includes the PDSCH.
[0176] In some examples, a modulation scheme of the WUS, a data rate of a symbol used to transmit the WUS, a waveform pattern of the WUS, or a combination thereof, indicate whether the SPS occasion includes the PDSCH.
[0177] In some examples, the WUS is a low-power WUS.
[0178] FIG. **11** shows a diagram of a system **1100** including a device **1105** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The device **1105** may be an example of or include components of a device **805**, a device **905**, or a network entity **105** as described herein. The device **1105** may communicate with other network devices or network equipment such as one or more of the network entities **105**, UEs **115**, or any combination thereof. The communications may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The device **1105** may include components that support outputting and obtaining communications, such as a communications manager **1120**, a transceiver **1110**, one or more antennas **1115**, at least one memory **1125**, code **1130**, and at least one processor **1135**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **1140**).
[0179] The transceiver **1110** may support bi-directional communications via wired links, wireless links, or both as described herein. In some examples, the transceiver **1110** may include a wired transceiver and may communicate bi-directionally with another wired transceiver. Additionally, or alternatively, in some examples, the transceiver **1110** may include a wireless transceiver and may communicate bi-directionally with another wireless transceiver. In some examples, the device **1105** may include one or more antennas **1115**, which may be capable of transmitting or receiving wireless transmissions (e.g., concurrently). The transceiver **1110** may also include a modem to modulate signals, to provide the modulated signals for transmission (e.g., by one or more antennas **1115**, by a wired transmitter), to receive modulated signals (e.g., from one or more antennas **1115**, from a wired receiver), and to demodulate signals. In some implementations, the transceiver **1110**

may include one or more interfaces, such as one or more interfaces coupled with the one or more antennas **1115** that are configured to support various receiving or obtaining operations, or one or more interfaces coupled with the one or more antennas **1115** that are configured to support various transmitting or outputting operations, or a combination thereof. In some implementations, the transceiver **1110** may include or be configured for coupling with one or more processors or one or more memory components that are operable to perform or support operations based on received or obtained information or signals, or to generate information or other signals for transmission or other outputting, or any combination thereof. In some implementations, the transceiver **1110**, or the transceiver **1110** and the one or more antennas **1115**, or the transceiver **1110** and the one or more antennas **1115** and one or more processors or one or more memory components (e.g., the at least one processor **1135**, the at least one memory **1125**, or both), may be included in a chip or chip assembly that is installed in the device **1105**. In some examples, the transceiver **1110** may be operable to support communications via one or more communications links (e.g., communication link(s) **125**, backhaul communication link(s) **120**, a midhaul communication link **162**, a fronthaul communication link **168**).

[0180] The at least one memory **1125** may include RAM, ROM, or any combination thereof. The at least one memory **1125** may store computer-readable, computer-executable, or processor-executable code, such as the code **1130**. The code **1130** may include instructions that, when executed by one or more of the at least one processor **1135**, cause the device **1105** to perform various functions described herein. The code **1130** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **1130** may not be directly executable by a processor of the at least one processor **1135** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the at least one memory **1125** may include, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some examples, the at least one processor **1135** may include multiple processors and the at least one memory **1125** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories which may, individually or collectively, be configured to perform various functions herein (for example, as part of a processing system).

[0181] The at least one processor **1135** may include one or more intelligent hardware devices (e.g., one or more general-purpose processors, one or more DSPs, one or more central processing units (CPUs), one or more graphics processing units (GPUs), one or more neural processing units (NPUs) (also referred to as neural network processors or deep learning processors (DLPs)), one or more microcontrollers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some cases, the at least one processor **1135** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into one or more of the at least one processor **1135**. The at least one processor **1135** may be configured to execute computer-readable instructions stored in a memory (e.g., one or more of the at least one memory **1125**) to cause the device **1105** to perform various functions (e.g., functions or tasks supporting techniques to activate SPS occasions). For example, the device **1105** or a component of the device **1105** may include at least one processor **1135** and at least one memory **1125** coupled with one or more of the at least one processor **1135**, the at least one processor **1135** and the at least one memory **1125** configured to perform various functions described herein. The at least one processor **1135** may be an example of a cloud-computing platform (e.g., one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (e.g., by executing code **1130**) to perform the functions of the device **1105**. The at least one processor **1135** may be any one or more suitable processors capable of executing scripts or instructions of one or more software programs stored in the device **1105** (such as within one or more of the at least one memory **1125**).

In some examples, the at least one processor **1135** may include multiple processors and the at least one memory **1125** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions herein. In some examples, the at least one processor **1135** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **1135**) and memory circuitry (which may include the at least one memory **1125**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **1135** or a processing system including the at least one processor **1135** may be configured to, configurable to, or operable to cause the device **1105** to perform one or more of the functions described herein. Further, as described herein, being "configured to," being "configurable to," and being "operable to" may be used interchangeably and may be associated with a capability, when executing code stored in the at least one memory **1125** or otherwise, to perform one or more of the functions described herein.

[0182] In some examples, a bus **1140** may support communications of (e.g., within) a protocol layer of a protocol stack. In some examples, a bus **1140** may support communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack), which may include communications performed within a component of the device **1105**, or between different components of the device **1105** that may be co-located or located in different locations (e.g., where the device **1105** may refer to a system in which one or more of the communications manager **1120**, the transceiver **1110**, the at least one memory **1125**, the code **1130**, and the at least one processor **1135** may be located in one of the different components or divided between different components).

[0183] In some examples, the communications manager **1120** may manage aspects of communications with a core network **130** (e.g., via one or more wired or wireless backhaul links). For example, the communications manager **1120** may manage the transfer of data communications for client devices, such as one or more UEs **115**. In some examples, the communications manager **1120** may manage communications with one or more other network entities **105**, and may include a controller or scheduler for controlling communications with UEs **115** (e.g., in cooperation with the one or more other network devices). In some examples, the communications manager **1120** may support an X**2** interface within an LTE/LTE-A wireless communications network technology to provide communication between network entities **105**.

[0184] The communications manager **1120** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **1120** is capable of, configured to, or operable to support a means for transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The communications manager **1120** is capable of, configured to, or operable to support a means for transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0185] By including or configuring the communications manager **1120** in accordance with examples as described herein, the device **1105** may support techniques for indicating empty SPS occasions via a WUS, which may provide for reduced processing, reduced power consumption, and a more efficient utilization of communication resources.

[0186] In some examples, the communications manager **1120** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the transceiver **1110**, the one or more antennas **1115** (e.g., where applicable), or any combination thereof. Although the communications manager **1120** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **1120** may be supported by or performed by the transceiver **1110**, one or more of the at least one processor **1135**, one or more of the at least one memory **1125**, the code **1130**, or any combination thereof (for example, by a processing system including at least a portion of the at least one processor **1135**, the at least one memory **1125**, the code **1130**, or any combination thereof). For example, the code **1130** may include instructions executable by one or more of the at least one processor **1135** to cause the device **1105** to perform various aspects of techniques to activate SPS occasions as described herein, or the at least one processor **1135** and the at least one memory **1125** may be otherwise configured to, individually or collectively, perform or support such operations.

[0187] FIG. **12** shows a flowchart illustrating a method **1200** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The operations of the method **1200** may be implemented by a UE or its components as described herein. For example, the operations of the method **1200** may be performed by a UE **115** as described with reference to FIGS. **1** through **7**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0188] At **1205**, the method may include receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The operations of **1205** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1205** may be performed by a control signaling component **625** as described with reference to FIG. **6**.

[0189] At **1210**, the method may include monitoring, via a second radio at the UE, for the WUS in accordance with the configuration. The operations of **1210** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1210** may be performed by a WUR component **630** as described with reference to FIG. **6**.

[0190] At **1215**, the method may include performing one of a first procedure or a second procedure based on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode. The operations of **1215** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1215** may be performed by an SPS procedure component **635** as described with reference to FIG. **6**.

[0191] FIG. **13** shows a flowchart illustrating a method **1300** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The operations of the method **1300** may be implemented by a UE or its components as described herein. For example, the operations of the method **1300** may be performed by a UE **115** as described with reference to FIGS. **1** through **7**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0192] At **1305**, the method may include receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The operations of **1305** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1305** may be performed by a control signaling component **625** as described with reference to FIG. **6**.

[0193] At **1310**, the method may include monitoring, via a second radio at the UE, for the WUS in accordance with the configuration, where the WUS indicates that the SPS occasion does not include a PDSCH. The operations of **1310** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1310** may be performed by a WUR component **630** as described with reference to FIG. **6**.

[0194] At **1315**, the method may include refraining from waking up the first radio of the UE based on the WUS indicating that the SPS occasion does not include the PDSCH. The operations of **1315** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1315** may be performed by a main radio component **640** as described with reference to FIG. **6**.

[0195] FIG. **14** shows a flowchart illustrating a method **1400** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The operations of the method **1400** may be implemented by a network entity or its components as described herein. For example, the operations of the method **1400** may be performed by a network entity as described with reference to FIGS. **1** through **3** and **8** through **11**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

[0196] At **1405**, the method may include transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The operations of **1405** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1405** may be performed by a scheduling component **1025** as described with reference to FIG. **10**.

[0197] At **1410**, the method may include transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH. The operations of **1410** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1410** may be performed by a WUS transmission component **1030** as described with reference to FIG. **10**.

[0198] FIG. **15** shows a flowchart illustrating a method **1500** that supports techniques to activate SPS occasions in accordance with one or more aspects of the present disclosure. The operations of the method **1500** may be implemented by a network entity or its components as described herein. For example, the operations of the method **1500** may be performed by a network entity as described with reference to FIGS. **1** through **3** and **8** through **11**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

[0199] At **1505**, the method may include transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion. The operations of **1505** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1505** may be performed by a scheduling component **1025** as described with reference to FIG. **10**.

[0200] At **1510**, the method may include transmitting the WUS in accordance with the configuration, where the WUS indicates to a UE whether the SPS occasion includes a PDSCH. The operations of **1510** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1510** may be performed by a WUS transmission component **1030** as described with reference to FIG. **10**.

[0201] At **1515**, the method may include transmitting, via the SPS occasion, the PDSCH. The operations of **1515** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1515** may be performed by an PDSCH transmission component **1035** as described with reference to FIG. **10**.

[0202] The following provides an overview of aspects of the present disclosure:

[0203] Aspect 1: A method for wireless communications at a UE, comprising: receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion; monitoring, via a second radio at the UE, for the WUS in accordance with the configuration; and performing one of a first procedure or a second procedure based at least in part on the monitoring, the first procedure pertaining to monitoring, via the first radio, the SPS occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

[0204] Aspect 2: The method of aspect 1, wherein the UE performs the first procedure based at least in part on the WUS indicating that the SPS occasion includes a PDSCH, wherein performing the first procedure comprises: waking up the first radio of the UE; monitoring, based at least in part on waking up the first radio, the SPS occasion; and receiving, via the first radio, the PDSCH based at least in part on the monitoring.

[0205] Aspect 3: The method of aspect 2, further comprising: monitoring, prior to receiving the PDSCH, for DCI based at least in part on the WUS indicating that the SPS occasion includes the PDSCH, the DCI indicating a set of parameters for receiving the PDSCH.

[0206] Aspect 4: The method of aspect 3, wherein the set of parameters comprise a quantity of RBs of the SPS occasion, a MCS associated with the PDSCH, a set of resources for indicating feedback associated with the PDSCH, or a combination thereof.

[0207] Aspect 5: The method of any of aspects 2 through 4, further comprising: refraining from monitoring for DCI based at least in part on the WUS further indicating for the UE to skip monitoring for the DCI, wherein receiving the PDSCH comprises: receiving the PDSCH via the SPS occasion according to a set of parameters indicated in the control signaling.

[0208] Aspect 6: The method of aspect 5, wherein the WUS indicating for the UE to skip monitoring for the DCI is based at least in part on a stability of a channel between the UE and a network entity, a variability of traffic load communicated via the channel, or both.

[0209] Aspect 7: The method of any of aspects 2 through 6, wherein the PDSCH carries data associated with XR applications.

[0210] Aspect 8: The method of any of aspects 2 through 7, wherein the PDSCH is a retransmission and the UE monitors for the retransmission or for a DCI that schedules the retransmission based at least in part on the WUS indicating that the SPS occasion includes the PDSCH.

[0211] Aspect 9: The method of any of aspects 1 through 8, wherein the UE performs the second procedure based at least in part on the WUS indicating that the SPS occasion does not include a PDSCH, wherein performing the second procedure comprises: refraining from waking up the first radio of the UE, wherein the first radio is maintained in the sleep mode based at least in part on refraining from waking up the first radio.

[0212] Aspect 10: The method of aspect 9, further comprising: refraining from transmitting feedback information associated with the PDSCH based at least in part on the WUS indicating that the SPS occasion does not include the PDSCH.

[0213] Aspect 11: The method of any of aspects 1 through 10, further comprising: performing the second procedure based at least in part on an absence of the WUS during resources associated with the WUS, wherein the absence of the WUS during the resources indicates that the SPS occasion does not include a PDSCH.

[0214] Aspect 12: The method of any of aspects 1 through 11, further comprising: performing the first procedure based at least in part on an absence of the WUS during resources associated with the WUS, wherein the absence of the WUS during the resources indicates that the SPS occasion includes a PDSCH.

[0215] Aspect 13: The method of any of aspects 1 through 12, wherein the control signaling further indicates a jitter range associated with the SPS occasion, and monitoring the WUS is in accordance with the jitter range associated with the SPS occasion.

[0216] Aspect 14: The method of any of aspects 1 through 13, wherein performance of the first procedure or the second procedure is based at least in part on bit values of one or more bits of the WUS.

[0217] Aspect 15: The method of any of aspects 1 through 14, wherein performance of the first procedure or the second procedure is based at least in part on a modulation scheme of the WUS, a data rate of a symbol used to transmit the WUS, a waveform pattern of the WUS, or a combination thereof.

[0218] Aspect 16: The method of any of aspects 1 through 15, wherein a periodicity of the WUS is based at least in part on a periodicity of the SPS occasion.

[0219] Aspect 17: The method of any of aspects 1 through 16, wherein the WUS is a low-power WUS.

[0220] Aspect 18: A method for wireless communications at a network entity comprising: transmitting control signaling that indicates a configuration associated with a WUS and that also indicates an association between the WUS and a SPS occasion; and transmitting the WUS in accordance with the configuration, wherein the WUS indicates to a UE whether the SPS occasion includes a PDSCH.

[0221] Aspect 19: The method of aspect 18, wherein the WUS indicates that the SPS occasion includes the PDSCH, the method further comprising: transmitting, via the SPS occasion, the PDSCH.

[0222] Aspect 20: The method of aspect 19, further comprising: transmitting, prior to transmitting the PDSCH, DCI that indicates a set of parameters associated with receiving the PDSCH.

[0223] Aspect 21: The method of aspect 20, wherein the set of parameters comprise a quantity of RBs of the SPS occasion, a MCS associated with the PDSCH, a set of resources for indicating feedback associated with the PDSCH, or a combination thereof.

[0224] Aspect 22: The method of any of aspects 19 through 21, further comprising: refraining from transmitting DCI based at least in part on the WUS further indicating for the UE to skip monitoring for the DCI.

[0225] Aspect 23: The method of aspect 22, wherein the WUS indicates for the UE to skip monitoring for the DCI is based at least in part on a stability of a channel between the UE and the network entity, a variability of traffic load communicated via the channel, or both.

[0226] Aspect 24: The method of any of aspects 19 through 23, wherein the PDSCH carries data associated with XR applications.

[0227] Aspect 25: The method of any of aspects 18 through 24, wherein the WUS indicates that the SPS occasion does not include the PDSCH, the method further comprising: refraining from transmitting the PDSCH via the SPS occasion.

[0228] Aspect 26: The method of any of aspects 18 through 25, further comprising: transmitting the WUS according to a periodicity that is based at least in part on a periodicity of the SPS occasion.

[0229] Aspect 27: The method of any of aspects 18 through 26, wherein the control signaling further indicates a jitter range associated with the SPS occasion, and transmitting the WUS is in accordance with the jitter range associated with the SPS occasion.

[0230] Aspect 28: The method of any of aspects 18 through 27, wherein bit values of one or more bits of the WUS indicate whether the SPS occasion includes the PDSCH.

[0231] Aspect 29: The method of any of aspects 18 through 28, wherein a modulation scheme of the WUS, a data rate of a symbol used to transmit the WUS, a waveform pattern of the WUS, or a combination thereof, indicate whether the SPS occasion includes the PDSCH.

[0232] Aspect 30: The method of any of aspects 18 through 29, wherein the WUS is a low-power WUS.

[0233] Aspect 31: A UE for wireless communications, comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the UE to perform a method of any of aspects 1 through 17.

[0234] Aspect 32: A UE for wireless communications, comprising at least one means for performing a method of any of aspects 1 through 17.

[0235] Aspect 33: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by one or more processors to perform a method of any of aspects 1 through 17.

[0236] Aspect 34: An apparatus for wireless communications, comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the apparatus to perform a method of any of aspects 18 through 30.

[0237] Aspect 35: An apparatus for wireless communications, comprising at least one means for performing a method of any of aspects 18 through 30.

[0238] Aspect 36: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by one or more processors to perform a method of any of aspects 18 through 30.

[0239] It should be noted that the methods described herein describe possible implementations. The operations and the steps may be rearranged or otherwise modified and other implementations are possible. Further, aspects from two or more of the methods may be combined.

[0240] Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.

[0241] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0242] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed using a general-purpose processor, a DSP, an ASIC, a CPU, a graphics processing unit (GPU), a neural processing unit (NPU), an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor but, in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). Any functions or operations described herein as being capable of being performed by a processor may be performed by multiple processors that, individually or collectively, are capable of performing the described functions or operations.

[0243] The functions described herein may be implemented using hardware, software executed by a processor, firmware, or any combination thereof. If implemented using software executed by a processor, the functions may be stored as or transmitted using one or more instructions or code of a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

[0244] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one location to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc. Disks may reproduce data magnetically, and discs may reproduce data optically using lasers. Combinations of the above are also included within the scope of computer-readable media. Any functions or operations described herein as being capable of being performed by a memory may be performed by multiple memories that, individually or collectively, are capable of performing the described functions or operations.

[0245] As used herein, including in the claims, "or" as used in a list of items (e.g., a list of items prefaced by a phrase such as "at least one of" or "one or more of") indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase "based on" shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as "based on condition A" may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase "based on" shall be construed in the same manner as the phrase "based at least in part on."

[0246] As used herein, including in the claims, the article "a" before a noun is open-ended and understood to refer to "at least one" of those nouns or "one or more" of those nouns. Thus, the terms "a," "at least one," "one or more," and "at least one of one or more" may be interchangeable. For example, if a claim recites "a component" that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. Thus, the term "a component" having characteristics or performing functions may refer to "at least one of one or more components" having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article "a" using the terms "the" or "said" may refer to any or all of the one or more components. For example, a component introduced with the article "a" may be understood to mean "one or more components," and referring to "the component" subsequently in the claims may be understood to be equivalent to referring to "at least one of the one or more components." Similarly, subsequent reference to a component introduced as "one or more components" using the terms "the" or "said" may refer to any or all of the one or more components. For example, referring to "the one or more components" subsequently in the claims may be understood to be equivalent to referring to "at least one of the one or more components."

[0247] The term "determine" or "determining" encompasses a variety of actions and, therefore, "determining" can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database, or another data structure), ascertaining, and the like. Also, "determining" can include receiving (e.g., receiving information), accessing (e.g., accessing data stored in memory), and the like. Also, "determining" can include resolving, obtaining, selecting, choosing, establishing, and other such similar actions.

[0248] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label or other

subsequent reference label.
[0249] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “example” used herein means “serving as an example, instance, or illustration” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some figures, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.
[0250] The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A user equipment (UE), comprising: one or more memories storing processor-executable code; and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the UE to: receive, via a first radio at the UE, control signaling that indicates a configuration associated with a wake-up signal and that also indicates an association between the wake-up signal and a semi-persistent scheduling occasion; monitor, via a second radio at the UE, for the wake-up signal in accordance with the configuration; and perform one of a first procedure or a second procedure based at least in part on the monitoring, the first procedure pertaining to monitoring, via the first radio, the semi-persistent scheduling occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.
2. The UE of claim 1, wherein the UE performs the first procedure based at least in part on the wake-up signal indicating that the semi-persistent scheduling occasion includes a physical downlink shared channel, and wherein, to perform the first procedure, the one or more processors are individually or collectively operable to execute the code to cause the UE to: wake up the first radio of the UE; monitor, based at least in part on waking up the first radio, the semi-persistent scheduling occasion; and receive, via the first radio, the physical downlink shared channel based at least in part on the monitoring.
3. The UE of claim 2, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to: monitor, prior to receiving the physical downlink shared channel, for downlink control information based at least in part on the wake-up signal indicating that the semi-persistent scheduling occasion includes the physical downlink shared channel, the downlink control information indicating a set of parameters for receiving the physical downlink shared channel.
4. The UE of claim 3, wherein the set of parameters comprise a quantity of resource blocks of the semi-persistent scheduling occasion, a modulation and coding scheme associated with the physical downlink shared channel, a set of resources for indicating feedback associated with the physical downlink shared channel, or a combination thereof.
5. The UE of claim 2, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to: refrain from monitoring for downlink control information based at least in part on the wake-up signal further indicating for the UE to skip monitoring for the downlink control information, wherein receiving the physical downlink shared channel comprises: receive the physical downlink shared channel via the semi-persistent

scheduling occasion according to a set of parameters indicated in the control signaling.
6. The UE of claim 5, wherein the wake-up signal indicating for the UE to skip monitoring for the downlink control information is based at least in part on a stability of a channel between the UE and a network entity, a variability of traffic load communicated via the channel, or both.
7. The UE of claim 2, wherein: the physical downlink shared channel carries data associated with extended reality applications.
8. The UE of claim 2, wherein the physical downlink shared channel is a retransmission and the UE monitors for the retransmission or for a downlink control information that schedules the retransmission based at least in part on the wake-up signal indicating that the semi-persistent scheduling occasion includes the physical downlink shared channel.
9. The UE of claim 1, wherein, to perform the second procedure, the one or more processors are individually or collectively operable to execute the code to cause the UE to: refrain from waking up the first radio of the UE, wherein the first radio is maintained in the sleep mode based at least in part on refraining from waking up the first radio.
10. The UE of claim 9, wherein the UE performs the second procedure based at least in part on the wake-up signal indicating that the semi-persistent scheduling occasion does not include a physical downlink shared channel, and wherein, to perform the second procedure, the one or more processors are individually or collectively operable to execute the code to cause the UE to: refrain from transmitting feedback information associated with the physical downlink shared channel based at least in part on the wake-up signal indicating that the semi-persistent scheduling occasion does not include the physical downlink shared channel.
11. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to: perform the second procedure based at least in part on an absence of the wake-up signal during resources associated with the wake-up signal, wherein the absence of the wake-up signal during the resources indicates that the semi-persistent scheduling occasion does not include a physical downlink shared channel.
12. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to: perform the first procedure based at least in part on an absence of the wake-up signal during resources associated with the wake-up signal, wherein the absence of the wake-up signal during the resources indicates that the semi-persistent scheduling occasion includes a physical downlink shared channel.
13. The UE of claim 1, wherein the control signaling further indicates a jitter range associated with the semi-persistent scheduling occasion, and monitoring the wake-up signal is in accordance with the jitter range associated with the semi-persistent scheduling occasion.
14. The UE of claim 1, wherein performance of the first procedure or the second procedure is based at least in part on bit values of one or more bits of the wake-up signal.
15. The UE of claim 1, wherein performance of the first procedure or the second procedure is based at least in part on a modulation scheme of the wake-up signal, a data rate of a symbol used to transmit the wake-up signal, a waveform pattern of the wake-up signal, or a combination thereof.
16. The UE of claim 1, wherein a periodicity of the wake-up signal is based at least in part on a periodicity of the semi-persistent scheduling occasion.
17. The UE of claim 1, wherein the wake-up signal is a low-power wake-up signal.
18. A network entity, comprising: one or more memories storing processor-executable code; and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the network entity to: transmit control signaling that indicates a configuration associated with a wake-up signal and that also indicates an association between the wake-up signal and a semi-persistent scheduling occasion; and transmit the wake-up signal in accordance with the configuration, wherein the wake-up signal indicates to a user equipment (UE) whether the semi-persistent scheduling occasion includes a physical downlink shared channel.
19. The network entity of claim 18, wherein the wake-up signal indicates that the semi-persistent

scheduling occasion includes the physical downlink shared channel, and the one or more processors are individually or collectively further operable to execute the code to cause the network entity to: transmit, via the semi-persistent scheduling occasion, the physical downlink shared channel.

20. The network entity of claim 19, wherein the one or more processors are individually or collectively further operable to execute the code to cause the network entity to: transmit, prior to transmitting the physical downlink shared channel, downlink control information that indicates a set of parameters associated with receiving the physical downlink shared channel.

21. The network entity of claim 20, wherein the set of parameters comprise a quantity of resource blocks of the semi-persistent scheduling occasion, a modulation and coding scheme associated with the physical downlink shared channel, a set of resources for indicating feedback associated with the physical downlink shared channel, or a combination thereof.

22. The network entity of claim 19, wherein the one or more processors are individually or collectively further operable to execute the code to cause the network entity to: refrain from transmitting downlink control information based at least in part on the wake-up signal further indicating for the UE to skip monitoring for the downlink control information.

23. The network entity of claim 22, wherein the wake-up signal indicates for the UE to skip monitoring for the downlink control information is based at least in part on a stability of a channel between the UE and the network entity, a variability of traffic load communicated via the channel, or both.

24. The network entity of claim 19, wherein: the physical downlink shared channel carries data associated with extended reality applications.

25. The network entity of claim 18, wherein the wake-up signal indicates that the semi-persistent scheduling occasion does not include the physical downlink shared channel, and the one or more processors are individually or collectively further operable to execute the code to cause the network entity to: refrain from transmitting the physical downlink shared channel via the semi-persistent scheduling occasion.

26. The network entity of claim 18, wherein the one or more processors are individually or collectively further operable to execute the code to cause the network entity to: transmit the wake-up signal according to a periodicity that is based at least in part on a periodicity of the semi-persistent scheduling occasion.

27. The network entity of claim 18, wherein the control signaling further indicates a jitter range associated with the semi-persistent scheduling occasion, and transmitting the wake-up signal is in accordance with the jitter range associated with the semi-persistent scheduling occasion.

28. The network entity of claim 18, wherein bit values of one or more bits of the wake-up signal indicate whether the semi-persistent scheduling occasion includes the physical downlink shared channel.

29. A method for wireless communications at a user equipment (UE), comprising: receiving, via a first radio at the UE, control signaling that indicates a configuration associated with a wake-up signal and that also indicates an association between the wake-up signal and a semi-persistent scheduling occasion; monitoring, via a second radio at the UE, for the wake-up signal in accordance with the configuration; and performing one of a first procedure or a second procedure based at least in part on the monitoring, the first procedure pertaining to monitoring, via the first radio, the semi-persistent scheduling occasion, and the second procedure pertaining to maintaining the first radio of the UE in a sleep mode.

30. A method for wireless communications at a network entity comprising: transmitting control signaling that indicates a configuration associated with a wake-up signal and that also indicates an association between the wake-up signal and a semi-persistent scheduling occasion; and transmitting the wake-up signal in accordance with the configuration, wherein the wake-up signal indicates to a user equipment (UE) whether the semi-persistent scheduling occasion includes a physical downlink shared channel.